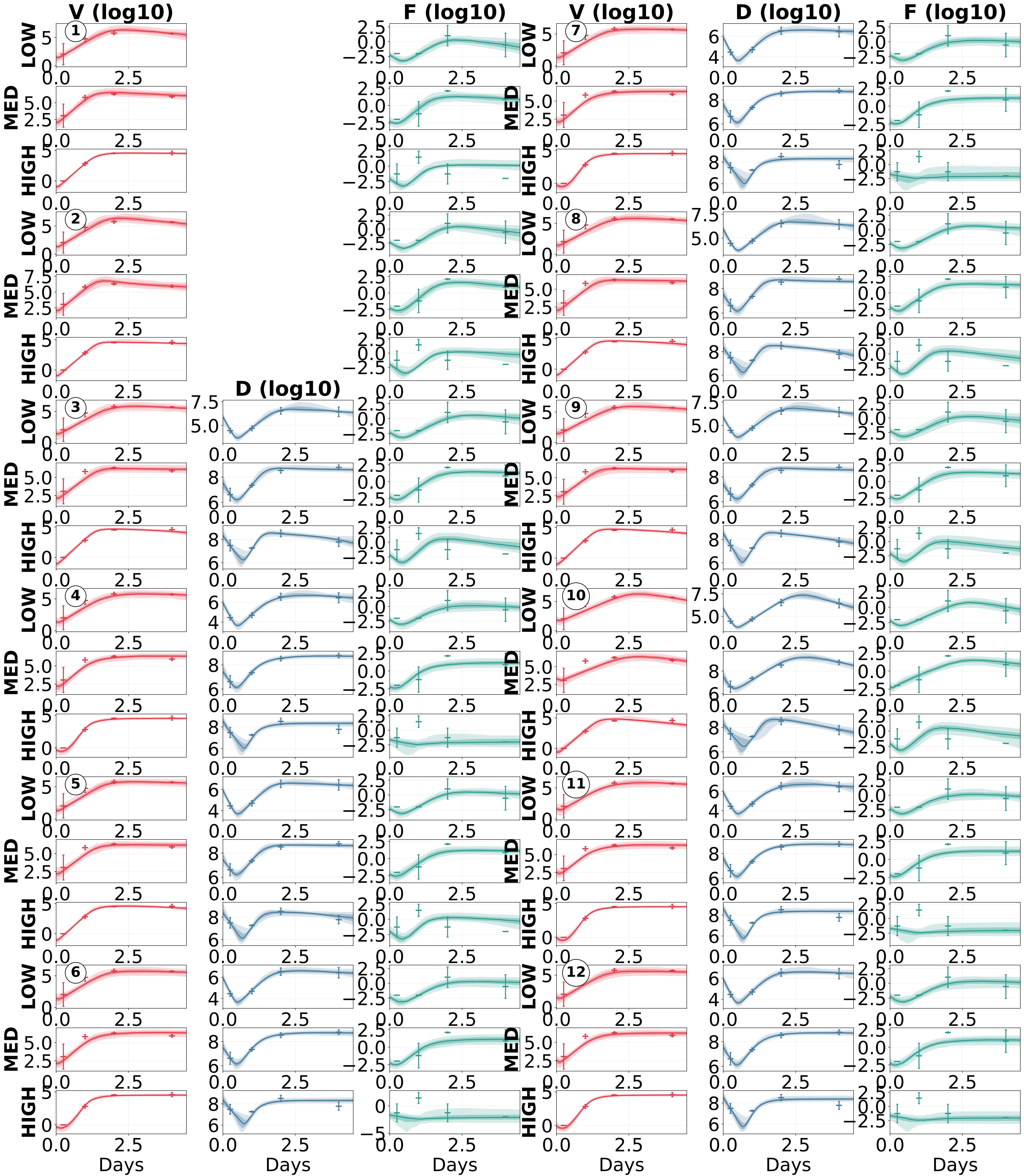




① TIVF e12:
 $k = k_{\text{hat}} / (1 + \epsilon_1 F)$
 $p_{\text{eff}} = p_{\text{hat}} / (1 + \epsilon_2 F)$
 $\frac{dT}{dt} = -k T V$
 $\frac{dI}{dt} = k T V - \delta I$
 $\frac{dV}{dt} = p_{\text{eff}} I - c V$
 $\frac{dF}{dt} = s_0 I - \alpha F$

② TIVF e2_delay:
 $p_{\text{eff}} = p_{\text{hat}} / (1 + \epsilon_2 F)$
 $\text{rate} = 5 / \tau$
 $\frac{dT}{dt} = -\beta T V$
 $\frac{dI}{dt} = \beta T V - \delta I$
 $\frac{dV}{dt} = p_{\text{eff}} I - c V$
 $\frac{dF}{dt} = s_0 Z_5 - \alpha F$
 $\frac{dZ_i}{dt} = \text{rate} (Z_{i-1} - Z_i), i=1..5$

③ TIVFD e2:
 $p_{\text{eff}} = p_{\text{hat}} / (1 + \epsilon_2 F)$
 $s(D) = s_0 + s_d D / (1 + \theta_s D)$
 $\frac{dT}{dt} = -\beta T V$
 $\frac{dI}{dt} = \beta T V - \delta I$
 $\frac{dV}{dt} = (1-d) p_{\text{eff}} I - c V$
 $\frac{dD}{dt} = d p_{\text{eff}} I - h D$
 $\frac{dF}{dt} = s(D) I - \alpha F$

④ TIVFD e3:
 $k_{\text{eff}} = k_{\text{hat}} / (1 + \epsilon_3 D)$
 $s(D) = s_0 + s_d D / (1 + \theta_s D)$
 $\frac{dT}{dt} = -k_{\text{eff}} T V$
 $\frac{dI}{dt} = k_{\text{eff}} T V - \delta I$
 $\frac{dV}{dt} = (1-d) p_{\text{eff}} I - c V$
 $\frac{dD}{dt} = d p_{\text{eff}} I - h D$
 $\frac{dF}{dt} = s(D) I - \alpha F$

⑤ TIVFD e12:
 $k = k_{\text{hat}} / (1 + \epsilon_1 F)$
 $p_{\text{eff}} = p_{\text{hat}} / (1 + \epsilon_2 F)$
 $s(D) = s_0 + s_d D / (1 + \theta_s D)$
 $\frac{dT}{dt} = -k T V$
 $\frac{dI}{dt} = k T V - \delta I$
 $\frac{dV}{dt} = (1-d) p_{\text{eff}} I - c V$
 $\frac{dD}{dt} = d p_{\text{eff}} I - h D$
 $\frac{dF}{dt} = s(D) I - \alpha F$

⑥ TIVFD e23:
 $k_{\text{eff}} = k_{\text{hat}} / (1 + \epsilon_3 D)$
 $p_{\text{eff}} = p_{\text{hat}} / (1 + \epsilon_2 F)$
 $s(D) = s_0 + s_d D / (1 + \theta_s D)$
 $\frac{dT}{dt} = -k_{\text{eff}} T V$
 $\frac{dI}{dt} = k_{\text{eff}} T V - \delta I$
 $\frac{dV}{dt} = (1-d) p_{\text{eff}} I - c V$
 $\frac{dD}{dt} = d p_{\text{eff}} I - h D$
 $\frac{dF}{dt} = s(D) I - \alpha F$

⑦ TIVFD e123:
 $k_{\text{eff}} = k_{\text{hat}} / (1 + \epsilon_1 F + \epsilon_3 D)$
 $p_{\text{eff}} = p_{\text{hat}} / (1 + \epsilon_2 F)$
 $s(D) = s_0 + s_d D / (1 + \theta_s D)$
 $\frac{dT}{dt} = -k_{\text{eff}} T V$
 $\frac{dI}{dt} = k_{\text{eff}} T V - \delta I$
 $\frac{dV}{dt} = (1-d) p_{\text{eff}} I - c V$
 $\frac{dD}{dt} = d p_{\text{eff}} I - h D$
 $\frac{dF}{dt} = s(D) I - \alpha F$

⑧ TIVFD e2_delay:
 $p_{\text{eff}} = p_{\text{hat}} / (1 + \epsilon_2 F)$
 $\text{rate} = 5 / \tau$
 $s(D_{\text{del}}) = s_0 + s_d D_{\text{del}} / (1 + \theta_s D_{\text{del}})$
 $\frac{dT}{dt} = -\beta T V$
 $\frac{dI}{dt} = \beta T V - \delta I$
 $\frac{dV}{dt} = (1-d) p_{\text{eff}} I - c V$
 $\frac{dD}{dt} = d p_{\text{eff}} I - h D$
 $\frac{dF}{dt} = s(D_{\text{del}}) I_{\text{del}} - \alpha F$

⑨ TIVFDR:
 $p_{\text{eff}} = p_{\text{hat}} / (1 + \epsilon_2 F)$
 $s(D) = s_d D / (1 + \theta_s D)$
 $\frac{dT}{dt} = -\beta T V - \phi F T + \rho R$
 $\frac{dR}{dt} = \phi F T - \rho R$
 $\frac{dI}{dt} = \beta T V - \delta I$
 $\frac{dV}{dt} = (1-d) p_{\text{eff}} I - c V$
 $\frac{dD}{dt} = d p_{\text{eff}} I - h D$
 $\frac{dF}{dt} = s(D) I - \alpha F$

⑩ TI1I2VFD:
 $s(D) = s_0 + s_d D / (1 + \theta_s D)$
 $\frac{dT}{dt} = -\beta T V$
 $\frac{dI_1}{dt} = \beta T V - k_{\text{ecl}} I_1$
 $\frac{dI_2}{dt} = k_{\text{ecl}} I_1 - \delta I_2$
 $\frac{dV}{dt} = (1-d) p_{\text{eff}} I_2 - c V$
 $\frac{dD}{dt} = d p_{\text{eff}} I_2 - h D$
 $\frac{dF}{dt} = s(D) I_2 - \alpha F$

⑪ TI1I2VFD e3:
 $\beta_{\text{eff}} = \beta_{\text{hat}} / (1 + \epsilon_3 D)$
 $s(D) = s_0 + s_d D / (1 + \theta_s D)$
 $\frac{dT}{dt} = -\beta_{\text{eff}} T V$
 $\frac{dI_1}{dt} = \beta_{\text{eff}} T V - k_{\text{ecl}} I_1$
 $\frac{dI_2}{dt} = k_{\text{ecl}} I_1 - \delta I_2$
 $\frac{dV}{dt} = (1-d) p_{\text{eff}} I_2 - c V$
 $\frac{dD}{dt} = d p_{\text{eff}} I_2 - h D$
 $\frac{dF}{dt} = s(D) I_2 - \alpha F$

⑫ TI1I2VFD e123:
 $\beta_{\text{eff}} = \beta_{\text{hat}} / (1 + \epsilon_3 D)$
 $k_{\text{eff}} = k_{\text{ecl}} / (1 + \epsilon_1 F)$
 $p_{\text{eff}} = p_7 / (1 + \epsilon_2 F)$
 $s(D) = s_0 + s_d D / (1 + \theta_s D)$
 $\frac{dT}{dt} = -\beta_{\text{eff}} T V$
 $\frac{dI_1}{dt} = \beta_{\text{eff}} T V - k_{\text{eff}} I_1$
 $\frac{dI_2}{dt} = k_{\text{eff}} I_1 - \delta I_2$
 $\frac{dV}{dt} = (1-d) p_{\text{eff}} I_2 - c V$
 $\frac{dD}{dt} = d p_{\text{eff}} I_2 - h D$
 $\frac{dF}{dt} = s(D) I_2 - \alpha F$